Omar Tabti

Engineering Student in Computer Science | Applied Mathematics & AI
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SUMMARY

Engineering student (**ENSEEIHT – INP Toulouse**, graduating 2029) in **Computer Science**, **High-Performance Computing & Big Data** track, coming from top-level preparatory classes (PCSI, PSI*) with solid foundations in **applied mathematics**, **numerical optimization** and **scientific computing**. Hands-on experience in numerical modelling and optimization through a research project (TIPE) on **Stirling engines** and **Flettner rotors**, and proficient in **Python**, **C**, **OCaml**, **Ada** and **SQL**. Seeking an internship in **artificial intelligence** and **quantitative analysis**.

EDUCATION

Engineering Degree – Computer Science, HPC & Big Data track – ***ENSEEIHT*** – *INP Toulouse* 2026 – 2029

- **Relevant coursework:** sparse linear algebra, numerical optimization, PDEs and scientific computing, operations research, graph theory, machine learning and statistics, concurrent systems and OpenMP, databases

Preparatory Classes – PCSI, then PSI* – *Lycée Déodat de Séverac, Toulouse* 2024 – 2026

PROFESSIONAL EXPERIENCE

Volunteer Tutor – Mathematics & Physics 2024 – 2026
Online community servers – Remote

- Helped final-year high-school students prepare for the final exam in **mathematics** and **physics**: exercise sessions, explanation of key concepts and exam methodology.
- Answered students' questions online on a regular basis and explained difficult topics clearly, strengthening my **communication** and **teaching** skills.

PROJECTS

Stirling Engine – Cycle Modelling & Optimization (TIPE) – *Lycée Déodat de Séverac – Preparatory Classes* 2025 – 2026

- Modelled the thermodynamic cycle of a **Stirling engine** and compared several **numerical integration** methods in terms of **accuracy**, and selected the most reliable one.
- Carried out the **optimization** of the engine parameters to maximize the efficiency with respect to the **Carnot efficiency**, using **Python**, and identified the parameter set giving the best efficiency.

Flettner Rotor – Performance Modelling & Optimization (TIPE) – *Lycée Déodat de Séverac – Preparatory Classes* 2024 – 2025

- Built a performance model of a **Flettner rotor** (**Magnus effect**) and optimized its **dimensions** to maximize efficiency, identifying the most favorable configuration.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, OCaml, Ada, SQL, MATLAB, JavaScript, HTML/CSS
- **Numerical Methods & Mathematics:** numerical optimization, numerical integration, sparse linear algebra, PDEs, statistics, operations research, graph theory
- **HPC & Systems (coursework):** OpenMP, concurrent programming, middleware, computer architecture
- **Data Science & Machine Learning (coursework):** machine learning, data analysis, databases (SQL)
- **Tools & Environment:** Linux, GitHub, Excel, Solidworks (CAD)
- **Soft Skills:** Efficiency, communication, teamwork, adaptability, continuous learning

LANGUAGES & INTERESTS

Languages: French (Native), Spanish (Native), Arabic (Native), Amazigh (Native), English (Advanced)
Interests: Football, AI & Tech, Mathematics, Computer Science, Travel